CONTINUOUS AND SMOOTH PLURISUBHARMONIC EXHAUSTIONS: THREEFOLD COUNTEREXAMPLES AND PARTIAL RESULTS FOR SURFACES

Abstract. Starting from Koike's modified Grauert example, we construct a smooth quasi-projective Kähler threefold with a locally Lipschitz plurisubharmonic exhaustion but no C^2 plurisubharmonic exhaustion. A modification over one point gives a non-Kähler threefold with the same properties. A separate, concise surface section establishes numerical boundary criteria for nonprojective compactifiers and smooth rulings, and records further affirmative projective and smooth-boundary cases. A classification table distinguishes these results from excluded configurations and cases not treated by the arguments. All compactification hypotheses and the recent elliptic-neighborhood input are stated explicitly. The unrestricted question in complex dimension two is not resolved.

1. Introduction and main results

The regularity of a plurisubharmonic exhaustion matters even for Kähler manifolds. This paper constructs a smooth quasi-projective threefold admitting a continuous plurisubharmonic exhaustion but no C^2 one, and then inserts a non-Kähler configuration over a single point without changing these exhaustion properties. In dimension two, the same line-bundle construction cannot give a counterexample. We prove this obstruction and several affirmative regularization results for surfaces, and specify the boundary problem left by our reductions. Throughout the introduction, "psh" abbreviates plurisubharmonic, and "smooth" means C^∞.

1.1. **Threefold counterexamples and their metric construction.** Our compact base is the modified Grauert ruled surface appearing in Koike's work [1, Example 4.1]. Its failure of semipositivity is already known; the first-order obstruction, including its C^2 version, is not claimed as a new result here [1, Theorem 1.4 and Example 4.1], [2, Remark 2.8]. We include a direct proof to make the regularity threshold used below explicit. The analytic construction then consists of a cusp correction, a global maximum, and a radius criterion converting any hypothetical C^2 exhaustion into a forbidden metric. Failure of smooth semipositivity by itself would not provide a continuous exhaustion.

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The cusp estimate also reaches the locally Lipschitz endpoint: a sufficiently large multiple of $q^{1/2}$ dominates the initial curvature in the scaled tangent variables. The coefficient is essential at this endpoint; for smaller exponents shrinking the neighborhood already suffices.

Theorem 1.1. *Let C be the smooth projective completion of*

$$y^2 = x^5 - x,$$

let $K = K_C$, and let

$$0 \longrightarrow K \longrightarrow V \longrightarrow \mathcal{O}_C \longrightarrow 0 \tag{1.1}$$

be a nonsplit holomorphic extension. Let

$$S = \mathbf{P}_{\text{lines}}(V), \qquad \pi : S \longrightarrow C, \qquad Y = \mathbf{P}(K) \subset S,$$

and set

$$L = \mathcal{O}_S(Y) \otimes \pi^* K, \qquad M = \operatorname{Tot}(L^{-1}). \tag{1.2}$$

Then:

(i) *S is a smooth projective surface and M is a smooth quasi-projective Kähler threefold;*
(ii) *L admits a continuous Hermitian metric with plurisubharmonic local weights, but admits no C^2 Hermitian metric with semipositive curvature;*
(iii) *M admits a continuous plurisubharmonic exhaustion, but admits no C^2 plurisubharmonic exhaustion.*

The metric and the exhaustion in (ii)–(iii) *can be chosen locally Lipschitz, and hence locally of class $C^{0,\gamma}$ for every $0 < \gamma \leq 1$. No assertion about existence of a C^1 exhaustion is made.*

The passage from metrics to exhaustions uses a criterion valid for a holomorphic line bundle A over any compact complex manifold Z:

$$\begin{gathered}\operatorname{Tot}(A^{-1}) \text{ admits a } C^r \text{ psh exhaustion} \\ \Longleftrightarrow \\ A \text{ admits a } C^r \text{ semipositive Hermitian metric}, \qquad r \in \{2, \infty\}.\end{gathered}$$

This is Corollary 3.4. A semipositive metric gives the squared dual norm exhaustion. Conversely, averaging an arbitrary exhaustion under fiberwise circle rotation and taking a sufficiently high level yields a radius function whose negative logarithm is a psh metric weight. Consequently, the obstruction in Theorem 1.1 rules out every C^2 psh exhaustion, not only a smoothing of the particular one constructed here.

The non-Kähler example has the same dimension and agrees with M away from a compact exceptional fiber.

Theorem 1.2. *For every point x of the threefold M in Theorem 1.1, there is a smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to M$ inducing a biholomorphism*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

The threefold N admits a locally Lipschitz psh exhaustion but no C^2 psh exhaustion.

The construction places Hironaka's opposite-order blowups inside the exceptional plane of the blowup of x. It produces two compact rational curves whose sum is zero in real homology, precluding a Kähler form. The exhaustion assertions follow from Proposition 4.1: a proper surjective holomorphic map that is biholomorphic away from a compact subset preserves the existence of C^r psh exhaustions, for $r = 0, 2, \infty$. The reverse implication uses convex composition to make an exhaustion constant near the exceptional set before descending it. Thus the modification is localized over one point; no small deformation of a fixed complex structure is asserted. Choosing x outside the zero section also preserves a neighborhood of the compact base. The compact product principle in Proposition 4.4 gives further examples in higher dimensions, including the Hopf-surface product in Corollary 4.5.

1.2. **Selected surface results and their scope.** The two-dimensional statements are collected in Section 5. For a line bundle A on a compact Riemann surface, its dual total space has a continuous psh exhaustion precisely when $\deg A \geq 0$, and then it has a smooth one (Proposition 5.2). This excludes a curve-base version of our threefold construction, not all possible surface counterexamples. For arbitrary noncompact surfaces with a nonconstant holomorphic function, regularization is a consequence of Vâjâitu's holomorphic-convexity theorem (Proposition 5.1).

The remaining geometric results assume a divisor compactification $X = S \setminus \operatorname{supp} D$. For nonprojective S, and for a smooth $\mathbf{P}^1$-bundle with reduced SNC boundary, continuous and smooth psh exhaustions are equivalent to the absence of a negative definite connected boundary component (Theorems 5.4 and 5.5). Further affirmative cases include disconnected projective boundary, suitable moving square-zero boundary divisors, torsion canonical bundle, and vertical boundaries of fibrations (Proposition 5.6). The torsion-canonical case is a standard application of surface log abundance. Proposition 5.8 gives additional sufficient conditions for smooth square-zero boundaries.

Table 1 separates what these arguments prove for compact minimal models from what they leave untreated. Passing to a minimal compactifier is not automatically compatible with the open complement. No classification of arbitrary noncompact surfaces, nor a general continuous-to-smooth regularization theorem in dimension two, is asserted. Sections 2–4 contain the threefold constructions; Section 5 is independent supplementary material.

1.3. **Relation to prior work and scope of the claims.** The known ruled-surface obstruction, Hironaka's opposite-order blowups, and the quoted

classification and Ueda theorems are inputs, not new constructions of this paper. The holomorphic-convexity consequence, the affine-line-bundle construction, and the use of surface abundance are attributed where used. The surface arguments combine these inputs with the boundary and character calculations; no priority claim is made for each intermediate criterion or for a general regularization theorem.

The self-contained threefold argument is the combination of the explicit bounded-potential metric and the radius obstruction, followed by a compactly supported modification. Remark 2.5 explains exactly which part of the bounded-potential question in [3, Section 4] this construction addresses. We make no claim that dimension three is the smallest possible dimension of a counterexample, nor that Lipschitz regularity is an optimal threshold. The full dimension-two problem is not settled in this article.

The elliptic-neighborhood step used in the surface section invokes Theorem 1.6 of the September 2026 preprint [14]. This input is independent of both threefold counterexamples. The classification table refers to the stated hypotheses, not to a purported complete classification of all continuously weakly complete surfaces.

1.4. **Conventions and terminology.** We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial.$$

For a holomorphic line bundle A and a local holomorphic frame e, a Hermitian metric is written

$$\|e\|_h^2 = e^{-\varphi}.$$

Its curvature is $\Theta_h(A) = dd^c\varphi$, representing $c_1(A)$. If $e_j = g_{ij}e_i$, the metric weights satisfy

$$\varphi_j = \varphi_i - \log|g_{ij}|^2. \tag{1.3}$$

A positive continuous Hermitian metric is semipositive in the current sense when its local weights are continuous psh functions. For a C^2 metric this agrees with pointwise semipositivity of its curvature.

For a C^2 real-valued function f, its unnormalized Levi form is

$$\mathcal{L}_f(\xi) = \sum_{j,k} f_{z_j\bar z_k}\xi_j\overline{\xi_k}.$$

Thus f is psh exactly when $\mathcal{L}_f \geq 0$. The fixed positive factor $1/(2\pi)$ in dd^c does not affect any positivity argument. Poincaré–Lelong takes the form

$$dd^c \log\|s_D\|_h^2 = [D] - \Theta_h(\mathcal{O}(D)), \tag{1.4}$$

where s_D is the canonical section of an effective divisor D.

Definition 1.3. A continuous real-valued function F on a complex manifold X is an *exhaustion* if every sublevel set $\{F \leq c\}$ is compact. Equivalently, $F(x_j) \to +\infty$ for every sequence escaping every compact subset of X.

We say that X is *continuously weakly complete* or *smoothly weakly complete* when it admits a continuous or smooth psh exhaustion, respectively. Specifying the regularity avoids the varying conventions for weak completeness in the literature. Our projectivization $\mathbf{P}_{\text{lines}}(E)$ parametrizes one-dimensional subspaces of the fibers of E, and its tautological line bundle is $\mathcal{O}_{\mathbf{P}(E)}(-1)$.

2. The ruled surface and continuous semipositive metrics

2.1. **The ruled surface and the metric obstruction.**

2.1.1. *The curve and the extension.* The polynomial $x^5 - x$ has five distinct roots. The double covering $C \to \mathbf{P}^1$ is therefore branched at these five roots and at infinity. Riemann–Hurwitz gives

$$2g(C) - 2 = 2(-2) + 6 = 2,$$

so $g(C) = 2$ and $\deg K = 2$. Serre duality gives

$$\operatorname{Ext}^1(\mathcal{O}_C, K) = H^1(C, K) \simeq H^0(C, \mathcal{O}_C)^* \simeq \mathbb{C}.$$

Consequently a nonzero extension class exists. All nonzero classes give isomorphic extensions after allowing scalar automorphisms of the end terms, so the isomorphism class of V is unambiguous.

One can also specify the extension through a Dolbeault operator. Choose a smooth positive $(1,1)$-form β on C with $\int_C \beta = 1$, and regard it as a K-valued $(0,1)$-form. On the smooth bundle $K \oplus \mathcal{O}_C$, set

$$\bar\partial_V(s, f) = (\bar\partial_K s + f\beta, \bar\partial f). \tag{2.1}$$

This operator is integrable, since there are no $(0,2)$-forms on a curve, and defines (1.1). A splitting would give a holomorphic lift $(s, 1)$ and hence $\bar\partial_K s = -\beta$. Since s is a smooth $(1,0)$-form, Stokes' theorem would then imply $\int_C \beta = 0$, a contradiction. The choice of the standard identification of K-valued $(0,1)$-forms with $(1,1)$-forms only changes this argument by a sign.

2.1.2. *The negative section and the line bundle.* By GAGA, V is algebraic, so $S = \mathbf{P}_{\text{lines}}(V)$ is a smooth projective surface. The inclusion $K \hookrightarrow V$ gives a section $Y = \mathbf{P}(K)$ of π. Its normal bundle is

$$N_{Y/S} \simeq \operatorname{Hom}(K, V/K) \simeq K^{-1}. \tag{2.2}$$

In particular,

$$\mathcal{O}_S(Y)|_Y \simeq K^{-1}, \qquad L|_Y \simeq \mathcal{O}_Y, \qquad Y^2 = -2. \tag{2.3}$$

The trivialization of $L|_Y$ in (2.3) is the canonical one induced by $K^{-1} \otimes K$.

For context, this line bundle is nef and big. If F is a fiber of π, then

$$L \cdot Y = 0, \qquad L \cdot F = 1, \qquad L^2 = 2.$$

For every irreducible curve $D \neq Y$ dominating C,

$$L \cdot D = Y \cdot D + 2\deg(\pi|_D) > 0.$$

Every vertical irreducible curve is a fiber. Hence L is nef, and the usual surface criterion $L^2 > 0$ implies bigness. For completeness, the latter also follows from Riemann–Roch: $h^2(S, L^m) = 0$ for m large, because $(K_S - mL)$ has negative intersection with a fixed ample divisor, and therefore

$$h^0(S, L^m) \geq \chi(S, L^m) = m^2 + O(m).$$

Moreover, $2L - Y = Y + 2\pi^* K$ is ample by Nakai–Moishezon: its square is 6, its intersections with Y and a fiber are 2 and 1, and its intersection with every other irreducible curve is positive.

2.1.3. *First-order transition functions.* Choose a finite coordinate cover $\{U_i\}$ of C, frames k_i of K, and holomorphic lifts v_i of 1 in (1.1). On overlaps,

$$k_j = g_{ij} k_i, \qquad v_j = v_i + c_{ij} k_i. \tag{2.4}$$

The cocycle $\{c_{ij} k_i\}$ represents the nonzero extension class. Near Y, write a line in V as

$$[k_i + w_i v_i].$$

Expressing this line in the j-th frame yields

$$w_j = \frac{g_{ij} w_i}{1 - c_{ij} w_i}. \tag{2.5}$$

Shrinking the neighborhoods if necessary, all denominators are nonzero. Let d_i be the frame of $\mathcal{O}_S(Y)$ defined by $s_Y = w_i d_i$. Then

$$\frac{d_j}{d_i} = \frac{w_i}{w_j} = \frac{1 - c_{ij} w_i}{g_{ij}}.$$

Thus the frames $\ell_i = d_i \otimes \pi^* k_i$ of L satisfy

$$\ell_j = (1 - c_{ij} w_i) \ell_i. \tag{2.6}$$

Their restrictions to Y agree, in accordance with the trivialization in (2.3).

Proposition 2.1. *The line bundle L admits no C^2 Hermitian metric with semipositive curvature.*

Proof. Suppose such a metric exists, with C^2 psh weights φ_i in the frames ℓ_i. By (2.6), the functions $\varphi_i|_Y$ agree on overlaps. They define a subharmonic function on the compact connected curve Y, hence a constant. Rescaling the metric, we may assume

$$\varphi_i(z, 0) = 0. \tag{2.7}$$

In particular $\varphi_{i, z\bar{z}}(z, 0) = 0$. The Levi matrix at $(z, 0)$ is positive semidefinite. A positive semidefinite Hermitian matrix with a zero diagonal entry has zero entries in the corresponding row and column. Therefore

$$\varphi_{i, w\bar{z}}(z, 0) = 0.$$

The C^1 function

$$a_i(z) = \varphi_{i, w}(z, 0)$$

satisfies the Cauchy–Riemann equation and is holomorphic.

Equations (1.3) and (2.6) give

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log|1 - c_{ij}(z)w_i|^2,$$

where the base coordinate change is understood. Differentiating with respect to w_i at zero and using (2.5), we obtain

$$g_{ij}a_j = a_i + c_{ij}. \tag{2.8}$$

The holomorphic lifts $v_i' = v_i - a_i k_i$ consequently agree on overlaps:

$$v_j' = v_i + c_{ij}k_i - a_j g_{ij}k_i = v_i - a_i k_i = v_i'.$$

They define a global holomorphic lift of 1, splitting (1.1). This is a contradiction. □

This calculation is the specialized first-order obstruction occurring in [1, Theorem 1.4 and Example 4.1]. Related obstruction classes, including their applicability to C^2 metrics, are discussed in [2]. Neither reference is needed for the preceding proof.

2.2. **A local continuous metric near a negative curve.** We isolate the analytic estimate used in the construction. The unitary flatness assumption means that a smooth metric of zero curvature exists on the restricted line bundle.

Lemma 2.2 (Cusp correction)**.** *Let D be a compact connected smooth curve in a complex surface X. Suppose $\deg N_{D/X} < 0$, and let A be a holomorphic line bundle such that $A|_D$ is unitary flat. Choose a smooth metric h_0 on A restricting to a flat metric on D, and a smooth metric h_D on $\mathcal{O}_X(D)$ whose curvature is strictly negative on TD. Put*

$$q = \|s_D\|^2_{h_D}.$$

For every $0 < a \le \frac{1}{2}$, there is a constant $\Lambda > 0$ such that the local weights of h_0 plus Λq^a are strictly psh off D on a sufficiently small neighborhood of D and extend as continuous psh weights across D. They are locally of class $C^{0,2a}$; in particular they are locally Lipschitz when $a = \frac{1}{2}$. For $0 < a < \frac{1}{2}$ one may take $\Lambda = 1$ after shrinking the neighborhood. At $a = \frac{1}{2}$, a sufficiently large Λ suffices.

Proof. The chosen metrics exist. Negative degree on the curve gives a negative curvature metric on $N_{D/X} = \mathcal{O}_X(D)|_D$. Any metric on a restricted line bundle extends smoothly to a neighborhood: extend the logarithm of its ratio to a fixed smooth metric. The same argument extends the flat metric on $A|_D$.

Use finitely many relatively compact coordinates (z, w) near D, with $D = \{w = 0\}$. Write

$$q = |w|^2 e^{-b(z,w)}, \qquad \beta = -b,$$

where b is the smooth weight of h_D. Negativity along TD says

$$\beta_{z\bar{z}}(z, 0) > 0. \tag{2.9}$$

For $w \neq 0$, set $\tau = \log q = \log|w|^2 + \beta$. The chain rule gives the exact identity

$$\mathcal{L}_{q^a}(v) = q^a\big(a\mathcal{L}_\tau(v) + a^2|\partial\tau(v)|^2\big). \tag{2.10}$$

Evaluate this on

$$v = \xi\frac{\partial}{\partial z} + w\nu\frac{\partial}{\partial w}.$$

Since $\log|w|^2$ is pluriharmonic off D,

$$\mathcal{L}_\tau(v) = \beta_{z\bar{z}}(z,0)|\xi|^2 + O(|w|)(|\xi|^2 + |\nu|^2),$$

and

$$\partial\tau(v) = \nu + \beta_z(z,0)\xi + O(|w|)(|\xi| + |\nu|).$$

Therefore

$$\begin{aligned}\frac{\mathcal{L}_{q^a}(\xi, w\nu)}{q^a} &= a\beta_{z\bar{z}}(z,0)|\xi|^2 + a^2|\nu + \beta_z(z,0)\xi|^2 \\ &\quad + O(|w|)(|\xi|^2 + |\nu|^2).\end{aligned} \tag{2.11}$$

All remainder bounds are uniform after choosing the finite cover and shrinking its coordinate neighborhoods.

The leading Hermitian form in (2.11) is positive definite: vanishing of its first term forces $\xi = 0$ by (2.9), and vanishing of the second then forces $\nu = 0$. Its smallest eigenvalue has a positive uniform lower bound on the compact coordinate pieces covering D. Also q^a is uniformly comparable to $|w|^{2a}$. Thus, after shrinking the neighborhoods,

$$\mathcal{L}_{q^a}(\xi, w\nu) \geq c|w|^{2a}(|\xi|^2 + |\nu|^2) \tag{2.12}$$

for a uniform $c > 0$.

Let φ_0 be a local weight of h_0. Its restriction to D has zero curvature, so $\varphi_{0,z\bar{z}}(z,0) = 0$. Smoothness gives

$$\varphi_{0,z\bar{z}} = O(|w|), \qquad \varphi_{0,z\bar{w}} = O(1), \qquad \varphi_{0,w\bar{w}} = O(1).$$

Consequently, in the same scaled variables,

$$|\mathcal{L}_{\varphi_0}(\xi, w\nu)| \leq C|w|(|\xi|^2 + |\nu|^2). \tag{2.13}$$

Combining (2.12) and (2.13) yields

$$\mathcal{L}_{\varphi_0 + \Lambda q^a}(\xi, w\nu) \geq \big(\Lambda c|w|^{2a} - C|w|\big)(|\xi|^2 + |\nu|^2). \tag{2.14}$$

If $0 < a < \frac{1}{2}$, take $\Lambda = 1$ and then shrink until the coefficient is positive. If $a = \frac{1}{2}$, fix $\Lambda > C/c$; the coefficient becomes $(\Lambda c - C)|w| > 0$ throughout the punctured neighborhood where the two uniform estimates hold. The constants c, C were obtained before choosing Λ, so this choice is not circular. Every tangent vector at $w \neq 0$ can be written in these scaled variables. This proves strict plurisubharmonicity off D in both cases.

The function $q^a = |w|^{2a}e^{-ab}$ is continuous and locally $C^{0,2a}$, including $2a = 1$. The local weights $\varphi_0 + \Lambda q^a$ are bounded near D and extend continuously to it. The removable singularity theorem for psh functions

across an analytic hypersurface therefore proves that these extended weights are psh. Adding the globally defined function Λq^a preserves the metric transition law (1.3). □

Remark 2.3. The scale in (2.11) is essential. The normal component of the Levi form has order $|w|^{2a-2}$, while the tangential component has order $|w|^{2a}$. These two scales allow the cusp to dominate the bounded mixed curvature of a smooth initial metric. At $a = \frac{1}{2}$ the scaled cusp and the initial error both have order $|w|$, which explains the coefficient choice in (2.14). Ordinary smoothing of the cusp would not preserve this domination at D. The argument proves no C^1 regularity.

2.3. **An explicit global continuous metric.** Return to the ruled surface S and the line bundle L in (1.2). Choose a positive smooth metric h_K on K. Using (2.3), extend the dual metric h_K^{-1} on $\mathcal{O}_S(Y)|_Y$ to a smooth metric h_Y on $\mathcal{O}_S(Y)$. Set

$$h_0 = h_Y \otimes \pi^* h_K, \qquad \theta_0 = \Theta_{h_0}(L), \qquad q = \|s_Y\|^2_{h_Y}. \tag{2.15}$$

The metric $h_0|_Y$ is the flat metric under the canonical trivialization of $L|_Y$, and

$$\theta_0|_{TY} = 0, \qquad \Theta_{h_Y}(\mathcal{O}(Y))|_{TY} = -\Theta_{h_K}(K) < 0. \tag{2.16}$$

Fix $0 < a \le \frac{1}{2}$ and choose $\Lambda > 0$ as in Lemma 2.2. Since q vanishes exactly on Y and S is compact, there exists $\varepsilon > 0$ such that the weights $\varphi_0 + \Lambda q^a$ are psh on $\{q < \varepsilon\}$. Write $Q = \max_S q$ and choose $0 < \varepsilon < Q$. Choose a constant

$$B > \max_{\varepsilon \le t \le Q} (\Lambda t^a - \log t). \tag{2.17}$$

Define, on $[0, Q]$,

$$\chi(0) = 0, \qquad \chi(t) = \max\{\Lambda t^a, \log t + B\} \quad (t > 0), \tag{2.18}$$

and put

$$h = h_0 \exp(-\chi(q)). \tag{2.19}$$

Proposition 2.4. *The metric (2.19) is a positive continuous Hermitian metric on L with psh local weights. It is locally $C^{0,2a}$, and is locally Lipschitz for $a = \frac{1}{2}$.*

Proof. The logarithmic branch in (2.18) tends to $-\infty$ as $t \to 0$, so $\chi(t) = \Lambda t^a$ for all sufficiently small t. Hence χ is continuous at zero and bounded on $[0, Q]$.

On $S \setminus Y$, Poincaré–Lelong and (2.15) give

$$\theta_0 + dd^c \log q = \pi^* \Theta_{h_K}(K) \ge 0. \tag{2.20}$$

As a current on S, the same expression equals $\pi^*\Theta_{h_K}(K) + [Y]$. Thus $\varphi_0 + \log q + B$ is a singular psh weight of L.

On $\{0 < q < \varepsilon\}$, the weights of h are

$$\varphi_0 + \chi(q) = \max\{\varphi_0 + \Lambda q^a, \varphi_0 + \log q + B\},$$

a maximum of two psh functions. They are therefore psh. On $\{q \geq \varepsilon\}$, (2.17) makes the logarithmic branch strictly dominant. This domination also holds on a collar of $\{q = \varepsilon\}$, so no boundary gluing issue arises. The weights are psh there by (2.20). Near Y, the cusp branch is dominant, and Lemma 2.2 proves positivity across Y.

All the weights are finite and continuous. Near Y their only nonsmooth term is $\Lambda q^a = \Lambda |w|^{2a} e^{-ab}$, which is $C^{0,2a}$ for $0 < 2a \leq 1$. Away from Y, both branches are smooth, and their maximum is locally Lipschitz, hence locally $C^{0,2a}$. This proves the asserted regularity. □

Remark 2.5 (Bounded potentials and minimal singularities). The current

$$T = \theta_0 + dd^c \chi(q) \in c_1(L)$$

has locally Lipschitz potentials when $a = \frac{1}{2}$. It also has minimal singularities: if $T' = \theta_0 + dd^c v \geq 0$ is another positive current in $c_1(L)$, its quasi-psh potential v is bounded above on compact S, while $\chi(q)$ is bounded below. Hence

$$v \leq \chi(q) + C_v$$

for a constant C_v. This is precisely the comparison expressing that T is no more singular than T'. Here global potentials exist by the $\partial\bar{\partial}$-lemma on the projective surface.

For this modified Grauert example, Proposition 2.4 therefore gives an affirmative answer to the bounded-potential question in [3, Section 4] for the modified example. That question also mentions Grauert's original example; the present argument does not identify its line bundle with L and does not settle that other case. The metric is constructed here, not taken from that preprint. This explicit answer for L is not a statement about arbitrary nef and big line bundles, nor a claim that every minimal-singularity representative has continuous potentials.

3. Line-bundle exhaustions and the Kähler counterexample

3.1. **Dual norms and compact-group averaging.**

Lemma 3.1. *Let Z be a compact complex manifold and $A \to Z$ a holomorphic line bundle with a positive continuous Hermitian metric h having psh local weights. Then the squared dual norm*

$$\rho(v) = \|v\|^2_{h^{-1}} \qquad (v \in \mathrm{Tot}(A^{-1}))$$

is a continuous psh exhaustion. If the weights are locally $C^{0,\gamma}$ for $0 < \gamma \leq 1$, so is ρ.

Proof. For a local frame e_i of A, write a dual vector as $v = \lambda_i e_i^*$, and write $h(e_i, e_i) = e^{-\varphi_i}$. Then

$$\rho = |\lambda_i|^2 e^{\varphi_i}. \tag{3.1}$$

Off the zero section, $\log \rho = \log |\lambda_i|^2 + \varphi_i$ is psh. Composition with the increasing convex function exp makes ρ psh there. Formula (3.1) extends continuously by zero to the zero section and is locally bounded, so removability gives psh on the whole total space.

Fix a smooth Hermitian metric h_{ref} on A^{-1}. By compactness of Z and positivity of h, constants $c, C > 0$ satisfy

$$c\|v\|^2_{h_{\mathrm{ref}}} \le \rho(v) \le C\|v\|^2_{h_{\mathrm{ref}}}.$$

The closed disk bundle of any fixed radius over Z is compact. Every sublevel of ρ is a closed subset of such a disk bundle and is therefore compact. The Hölder assertion follows locally from (3.1). □

Lemma 3.2 (Averaging preserves exhaustions)**.** *Let a compact group G act continuously on a locally compact Hausdorff space X, and let F be a continuous exhaustion. Its normalized Haar average*

$$\overline{F}(x) = \int_G F(gx)\, dg$$

is also an exhaustion. If G is a compact Lie group, X is a complex manifold, the action is smooth and holomorphic, and F is C^2 psh, then $\overline{F}$ is C^2 psh. The same assertion holds in the smooth category.

Proof. For $K_c = \{F \le c\}$, the orbit saturation $G{\cdot}K_c$ is compact. If $x \notin G{\cdot}K_c$, then $F(gx) > c$ for every $g \in G$, and continuity on the compact orbit implies $\overline{F}(x) > c$. Hence

$$\{\overline{F} \le c\} \subset G \cdot K_c.$$

The sublevel on the left is closed and therefore compact. Differentiation under the integral preserves C^2 regularity, and averaging the nonnegative Levi forms of the holomorphic pullbacks preserves psh. □

3.2. **The radius criterion.**

Proposition 3.3 (The radius criterion)**.** *Let Z be a compact complex manifold and $A \to Z$ a holomorphic line bundle. If $\mathrm{Tot}(A^{-1})$ admits a C^2 psh exhaustion, then A admits a C^2 Hermitian metric with semipositive curvature. If the exhaustion is smooth, the metric is smooth.*

Proof. Average the exhaustion over the holomorphic circle action $v \mapsto e^{i\alpha}v$ on the fibers, using Lemma 3.2. Denote the resulting invariant exhaustion by F. For a local frame e_i of A and $v = \lambda_i e_i^*$, circle invariance gives

$$F(z, \lambda_i) = U_i(z, t), \qquad t = \log |\lambda_i|^2, \quad \lambda_i \neq 0. \tag{3.2}$$

The function

$$V_i(z, \zeta) = U_i(z, \operatorname{Re}\zeta) = F(z, e^{\zeta/2}) \tag{3.3}$$

is a holomorphic pullback of F and is psh. In particular $U_{i,tt} \ge 0$.

For fixed z, continuity at the zero section gives the finite limit

$$\lim_{t \to -\infty} U_i(z, t) = F(z, 0).$$

Exhaustiveness gives $U_i(z,t) \to +\infty$ as $t \to +\infty$. A differentiable convex function with finite left limit has nonnegative derivative. Moreover, if its derivative vanishes at t_0, convexity forces it to be constant on $(-\infty, t_0]$. Consequently

$$U_{i,t}(z,t) > 0 \quad \text{whenever } U_i(z,t) > F(z,0). \tag{3.4}$$

Fix $c > \max_Z F|_Z$. By convexity, the two limits, and (3.4), there is a unique solution

$$U_i(z, \sigma_i(z)) = c, \qquad U_{i,t}(z, \sigma_i(z)) > 0. \tag{3.5}$$

The implicit function theorem makes σ_i of class C^2, or smooth when F is smooth.

We verify the sign of its Levi form explicitly. Differentiating (3.5) twice in base coordinates gives

$$\begin{aligned} 0 = {} & U_{i,z_j\bar{z}_k} + U_{i,z_jt}\sigma_{i,\bar{z}_k} + U_{i,t\bar{z}_k}\sigma_{i,z_j} \\ & + U_{i,tt}\sigma_{i,z_j}\sigma_{i,\bar{z}_k} + U_{i,t}\sigma_{i,z_j\bar{z}_k}, \end{aligned} \tag{3.6}$$

with all derivatives of U_i evaluated at $(z, \sigma_i(z))$. For V_i in (3.3), its $\zeta\bar{\zeta}$ entry is $U_{i,tt}/4$, and its mixed entries have a factor $1/2$. Thus (3.6) is precisely the identity

$$\mathcal{L}_{V_i}(\xi, 2\partial\sigma_i(\xi)) = -U_{i,t}\mathcal{L}_{\sigma_i}(\xi). \tag{3.7}$$

The left side is nonnegative and $U_{i,t} > 0$, so $\psi_i = -\sigma_i$ is psh.

If $e_j = g_{ij}e_i$, the dual coordinates satisfy $\lambda_j = g_{ij}\lambda_i$. The level in (3.5) consequently transforms by

$$\sigma_j = \sigma_i + \log|g_{ij}|^2,$$

so

$$\psi_j = \psi_i - \log|g_{ij}|^2.$$

These are exactly the metric transition laws (1.3). Hence $\|e_i\|^2 = e^{-\psi_i}$ defines the desired semipositive Hermitian metric on A. □

Corollary 3.4. *For a holomorphic line bundle A over a compact complex manifold, $\mathrm{Tot}(A^{-1})$ admits a C^2 psh exhaustion if and only if A admits a C^2 semipositive Hermitian metric. The same equivalence holds with C^2 replaced by C^∞.*

Proof. The forward implication is Proposition 3.3. For the reverse implication, the squared dual norm in Lemma 3.1 has the regularity of the given metric, by its local expression (3.1). □

3.3. **Proof of the Kähler counterexample.**

Proof of Theorem 1.1. The construction in Subsection 2.1 gives a smooth projective surface S and an algebraic line bundle L. The total space M is the open subset

$$M = \mathbf{P}_{\text{lines}}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}_{\text{lines}}(L^{-1}). \tag{3.8}$$

The identification sends $v \in L_z^{-1}$ to the line $[(1, v)]$. The projective bundle in (3.8) is a smooth projective threefold. Restricting any of its Kähler forms proves that M is Kähler.

Let h be the continuous semipositive metric from Proposition 2.4. If $p : M \to S$ is the bundle projection, the exhaustion is explicitly

$$\rho(v) = \|v\|_{h^{-1}}^2 = \|v\|_{h_0^{-1}}^2 \exp\bigl(\chi(q(p(v)))\bigr). \tag{3.9}$$

Lemma 3.1 shows that ρ is continuous, psh, and exhaustive. Taking $a = \frac{1}{2}$ and the coefficient Λ supplied by Lemma 2.2 makes h, its weights, and ρ locally Lipschitz. In particular the same example has local $C^{0,\gamma}$ regularity for all $0 < \gamma \leq 1$.

If a C^2 psh exhaustion on M existed, Proposition 3.3 would produce a C^2 semipositive Hermitian metric on L. Proposition 2.1 excludes such a metric. Therefore no C^2 psh exhaustion exists, completing the proof. □

Remark 3.5 (Interpretation). The example separates existence of a continuous exhaustion from existence of any smooth exhaustion. The conclusion does not depend on whether the particular function (3.9) can be smoothed: every hypothetical C^2 exhaustion leads to the forbidden metric. The argument does not determine whether a C^1 psh exhaustion exists, and the surface results below do not establish that dimension three is minimal. Thus a definition of weak pseudoconvexity requiring a smooth psh exhaustion is strictly stronger than the corresponding definition requiring only a continuous psh exhaustion, even for smooth quasi-projective Kähler threefolds.

4. Compact modifications and non-Kähler counterexamples

The Kähler counterexample can be changed over a single point to obtain a non-Kähler counterexample of the same dimension. We first isolate the exhaustion argument, then insert Hironaka's opposite-order blowup configuration in an exceptional plane. Compact products give a simpler alternative when an increase in dimension is allowed.

4.1. **Exhaustions under compactly supported modifications.**

Proposition 4.1. *Let $f : X' \to X$ be a proper surjective holomorphic map between nonempty complex manifolds. Suppose that, for a compact set $K \subset X$, it restricts to a biholomorphism*

$$X' \setminus f^{-1}(K) \longrightarrow X \setminus K.$$

For every $r \in \{0, 2, \infty\}$, X admits a C^r psh exhaustion if and only if X' does.

Proof. The case $K = \varnothing$ is immediate. Holomorphic pullback preserves plurisubharmonicity and the stated regularity, and properness makes the sublevels of a pulled-back exhaustion compact. Conversely, suppose F is a

C^r psh exhaustion on X'. Choose a relatively compact open neighborhood U of K and

$$c > \max_{f^{-1}(\overline{U})} F.$$

Choose a smooth convex nondecreasing function $\chi : \mathbb{R} \to [0, \infty)$ that vanishes on $(-\infty, c]$ and tends to infinity. For example, integrate a smooth nondecreasing function which is zero up to c and one from $c+1$ onward. On $X \setminus K$ define

$$G = \chi \circ F \circ (f|_{X' \setminus f^{-1}(K)})^{-1}.$$

This function is C^r and psh. It vanishes on $U \setminus K$, so extension by zero on K gives a C^r psh function on all of X. For each nonempty sublevel $\{G \leq a\}$ choose b such that $\chi(t) \leq a$ implies $t \leq b$. Then

$$\{G \leq a\} \subset K \cup f(\{F \leq b\}).$$

The left side is closed and the right side is compact. Thus G is an exhaustion; no smooth descent of the original function F is needed. □

4.2. **A non-Kähler modification over one point.**

Proposition 4.2. *Let X be a connected smooth complex threefold and $x \in X$. There is a connected smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to X$ inducing a biholomorphism*

$$N \setminus \mu^{-1}(x) \simeq X \setminus \{x\}.$$

Proof. Blow up x, obtaining

$$b : W = \mathrm{Bl}_x X \longrightarrow X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In homogeneous coordinates $[u : v : w]$ on E, take

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}.$$

The line A and smooth conic B meet transversely at exactly $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Near either point they are the two coordinate axes in a smooth surface E inside W.

Over $W \setminus \{q\}$, first blow up $A \setminus \{q\}$ and then the strict transform of $B \setminus \{q\}$. Over $W \setminus \{p\}$, first blow up $B \setminus \{p\}$ and then the strict transform of $A \setminus \{p\}$. All these centers are closed smooth curves in the respective open manifolds. On $W \setminus \{p, q\}$ the centers are disjoint, so the blowups commute canonically over the base. Glue the two resulting manifolds through this isomorphism and write

$$\nu : N \longrightarrow W, \qquad \mu = b \circ \nu.$$

The glued space is Hausdorff: points with distinct images can be separated in W, and points with the same image belong to one of the Hausdorff open manifolds lying over the two base opens. It is smooth of dimension three because every blowup center is smooth. Properness is local on the target, so ν and μ are proper. Each blowup chart is connected and the charts overlap, hence N is connected. All centers lie in E, so μ is an isomorphism away

from x. This is Hironaka's opposite-order construction [16, Section 6]; see also [17, Tag 08J1].

For completeness, the obstruction to a Kähler form is entirely contained in compact exceptional surfaces. Let $D_A, D_B \subset N$ be the exceptional divisors dominating A, B, and let $\alpha, \beta \in H_2(N, \mathbb{R})$ be the classes of their general $\mathbf{P}^1$ fibers. The surfaces D_A, D_B are smooth and compact: locally each is a ruled surface or its blowup at a point, and they map properly to the compact curves A, B. The fiber divisors of each map are linearly equivalent, since $A \simeq B \simeq \mathbf{P}^1$, and hence are homologous on the corresponding compact surface.

Over p, the strict transform of B meets the first exceptional divisor transversely at one point. The second blowup therefore restricts on that divisor to a surface blowup at that point. Its fiber over p is the reduced union of the strict transform $R_p \simeq \mathbf{P}^1$ of the first exceptional fiber and the new exceptional fiber $T_p \simeq \mathbf{P}^1$. Both occur with multiplicity one. Indeed, take local coordinates (x, y, z) with $A = \{y = z = 0\}$ and $B = \{x = z = 0\}$. In the first blowup chart $z = yt$, the first exceptional divisor is $y = 0$ and the second center is $x = t = 0$. Its restriction to $y = 0$ is the blowup of the smooth surface with coordinates (x, t) at the origin. The pullback of the fiber equation $x = 0$ is its strict transform plus the exceptional curve, each once. The curve T_p is also the fiber of $D_B \to B$ over p. Thus

$$\alpha = [R_p] + [T_p], \qquad \beta = [T_p].$$

Over q the order is reversed. With analogous notation,

$$\beta = [R_q] + [T_q], \qquad \alpha = [T_q].$$

These relations hold in ordinary homology: they arise from fiber divisors on compact D_A and D_B, regardless of compactness of N. Adding them gives

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

If ω were a Kähler form on N, closedness and (4.1) would imply

$$0 = \int_{R_p} \omega + \int_{R_q} \omega > 0,$$

a contradiction. □

Proof of Theorem 1.2. Apply Proposition 4.2 to the threefold M of Theorem 1.1 at the chosen point x. If ρ is its continuous psh exhaustion, then

$$\rho_N = \rho \circ \mu, \qquad \{\rho_N \le a\} = \mu^{-1}(\{\rho \le a\})$$

is a continuous psh exhaustion of N. A holomorphic map is locally Lipschitz in smooth coordinate charts, so local Lipschitz regularity of the exhaustion is preserved. A C^2 psh exhaustion on N would give one on M by Proposition 4.1 with $K = \{x\}$, contrary to Theorem 1.1. □

Remark 4.3 (The sense in which the change is local). The point x can be chosen outside the zero section of $M \to S$. The construction then preserves a

neighborhood of that section and, by properness, a neighborhood of infinity. More precisely, for every neighborhood U of x, μ is an isomorphism over $M \setminus U$. This is a modification supported over one point; it makes no claim about a small deformation of the complex structure on a fixed smooth manifold.

4.3. **Compact products and an explicit fivefold.**

Proposition 4.4 (Compact product principle)**.** *Let X be a nonempty complex manifold and T a nonempty compact connected complex manifold. For each $r \in \{0, 2, \infty\}$, X admits a C^r psh exhaustion if and only if $X \times T$ does. If T is non-Kähler, then $X \times T$ is non-Kähler.*

Proof. Pullback by the projection preserves regularity and plurisubharmonicity, with compact sublevels

$$\{f \circ \mathrm{pr}_X \le a\} = \{f \le a\} \times T. \tag{4.2}$$

Conversely, restriction to the closed holomorphic slice $X \times \{t_0\}$ preserves regularity, plurisubharmonicity, and compact sublevels. A Kähler form on the product would restrict to one on $\{x_0\} \times T$, proving the final assertion. In fact, the maximum principle makes every continuous psh function on the product constant on each compact connected fiber $\{x\} \times T$. □

Corollary 4.5 (A non-Kähler fivefold)**.** *Let M be the threefold of Theorem 1.1 and let*

$$H = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle. \tag{4.3}$$

Then $M \times H$ is a smooth non-Kähler fivefold admitting a continuous psh exhaustion but no C^2 psh exhaustion. For every $0 < \gamma < 1$ its exhaustion can be chosen locally $C^{0,\gamma}$.

Proof. The dilation action is free and properly discontinuous: if a compact set lies in $r \le \|z\| \le R$, only finitely many integers n satisfy $r/R \le 2^n \le R/r$. Polar coordinates identify its quotient smoothly with

$$H \simeq S^3 \times (\mathbb{R}/(\log 2)\mathbb{Z}) \simeq S^3 \times S^1. \tag{4.4}$$

Thus H is a compact connected complex surface with $H^2_{\mathrm{dR}}(H) = 0$. Any Kähler form ω would be exact, and Stokes' theorem would give $\int_H \omega^2 = 0$, contradicting positivity. Apply Proposition 4.4 and Theorem 1.1. Explicitly,

$$\rho_{M \times H}(v, [z]) = \rho(v) \tag{4.5}$$

is the required exhaustion, with the same local Hölder regularity. □

5. Surface results: proved cases and remaining cases

The question in dimension two is whether a continuous psh exhaustion implies the existence of a C^2, or a smooth, psh exhaustion. All positive results below give a smooth exhaustion. We do not construct a surface counterexample. We distinguish statements about an arbitrary noncompact

surface X from statements requiring a compactification $X = S \setminus \operatorname{supp} D$, where S is a smooth compact connected complex surface and $D \neq 0$ is effective. The Enriques–Kodaira classification concerns the compact surface S, not X. In particular, a classification of compact minimal surfaces is not a regularization theorem for arbitrary open surfaces. Table 1 summarizes exactly the cases established here.

5.1. **Two basic affirmative cases.**

Proposition 5.1 (A known holomorphic-convexity consequence)**.** *Let X be a connected noncompact complex surface with a continuous psh exhaustion. If $\mathcal{O}(X) \neq \mathbb{C}$, then X has a smooth psh exhaustion.*

Proof. The holomorphic-convexity theorem of Vâjâitu [4, Theorem 1.1] makes X holomorphically convex. Its Remmert reduction $r : X \to Y$ is proper with Stein target. A smooth strongly psh exhaustion of the complex space Y pulls back to a smooth psh exhaustion of X. Here smoothness on Y means local restriction of smooth ambient functions, so the pullback is smooth even over singular points of Y. □

This is a consequence of the cited theorem, not a new holomorphic-convexity result. It leaves the case $\mathcal{O}(X) = \mathbb{C}$ untreated. Likewise, the regularization theorem of Mongodi–Slodkowski for domains in a surface already carrying a real-analytic psh exhaustion [5, Corollaries 4.4–4.5] requires that ambient hypothesis; one cannot assume it for X itself.

Proposition 5.2 (Line-bundle total spaces over a curve)**.** *Let A be a holomorphic line bundle on a compact connected Riemann surface B. The following are equivalent:*

$$\operatorname{Tot}(A^{-1}) \text{ has a continuous psh exhaustion}$$
$$\iff \quad \deg A \geq 0 \quad \iff$$
$$\operatorname{Tot}(A^{-1}) \text{ has a smooth psh exhaustion.}$$

Proof. Choose a smooth reference metric with curvature θ_0 and an area form ω with $\int_B \omega = 1$. For $d = \deg A \geq 0$, solve $dd^c f = d\omega - \theta_0$. The resulting metric has curvature $d\omega$, and its squared dual norm is a smooth psh exhaustion by Lemma 3.1. Conversely, the compactification

$$\operatorname{Tot}(A^{-1}) = \mathbf{P}_{\text{lines}}(\mathcal{O}_B \oplus A^{-1}) \setminus \mathbf{P}_{\text{lines}}(A^{-1})$$

has infinity section with normal bundle A. If $\deg A < 0$, that section has negative self-intersection, contradicting the boundary obstruction in Lemma 5.3(i) below. □

Thus the line-bundle construction of Theorem 1.1 cannot be lowered to dimension two by replacing its compact base by a curve. This does not exclude other surface constructions.

5.2. **Boundary tools and permitted modifications.** For a divisor complement, a continuous exhaustion tends uniformly to $+\infty$ at the boundary: each sublevel is a compact subset of X, hence its complement in S is a neighborhood of $\operatorname{supp} D$. We call a psh function on a punctured boundary neighborhood which diverges in this way a *divergent collar*.

Blowing up a boundary point preserves X only when the new boundary is the *full reduced inverse image*, including the exceptional curve. This permits resolution of the boundary without changing X. Blowups at interior points give proper modifications with compact exceptional locus, and Proposition 4.1 preserves both exhaustion properties. By contrast, contracting a curve which meets but is not contained in the boundary need not induce a proper map of the complements. We make no unrestricted reduction of projective pairs to minimal compactifiers.

Lemma 5.3 (Boundary obstruction and collar construction)**.** *Let S be smooth compact and let $D \neq 0$ be effective.*

(i) *If a connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix, then $S \setminus \operatorname{supp} D$ has no continuous psh exhaustion.*

(ii) *If every connected boundary component admits a smooth divergent psh collar, then $S \setminus \operatorname{supp} D$ has a smooth psh exhaustion.*

(iii) *Let $E = \bigcup E_i$ be a connected reduced simple normal crossing* (SNC) *divisor and let λ be the largest eigenvalue of $I = (E_i \cdot E_j)$. If $\lambda > 0$, E admits a smooth strictly psh divergent collar. If $\lambda = 0$, the matrix is negative semidefinite with one-dimensional kernel, and there are positive integers a_i with*

$$A = \sum_i a_i E_i, \qquad A \cdot E_i = 0, \qquad A^2 = 0. \tag{5.1}$$

When S is projective, this divisor A is nef.

Proof. For (i), Grauert's contraction theorem [6] contracts the indicated component to a point p of a normal surface, without changing a punctured neighborhood. A continuous psh function on such a punctured neighborhood is bounded above near p. Indeed, a local finite projection to a bidisc, with p the only point over 0, sends it to the fiberwise maximum v_*, a continuous psh function on the punctured bidisc. This maximum is psh off the branch locus; continuity and local boundedness extend this across the branch locus away from 0. On

$$\{|z_1| \leq r,\ |z_2| = r\}$$

it has a finite maximum. The maximum principle on each vertical disc with $z_1 \neq 0$, followed by continuity at $z_1 = 0$, gives the same bound on the punctured smaller bidisc. This contradicts divergence of an exhaustion at the contracted boundary.

For (ii), choose disjoint smaller boundary neighborhoods W_E compactly contained in the collar neighborhoods and choose T_E larger than the collar function on ∂W_E. A smooth convex nondecreasing function χ_E, zero on

$(-\infty, T_E]$ and tending to infinity, makes $\chi_E(\Phi_E)$ vanish near ∂W_E. Extend by zero outside W_E and sum over the finitely many boundary components. The result is smooth and psh, diverges at every boundary point, and is proper because S is compact.

For (iii), the off-diagonal entries of I are nonnegative and its graph is connected. Perron–Frobenius applied after a scalar shift makes λ simple with a strictly positive eigenvector. If $\lambda > 0$, rational approximation and clearing denominators give $A = \sum a_i E_i$ with $a_i > 0$ and $A \cdot E_i > 0$ for every i. We spell out the metric construction needed here. The restriction $\mathcal{O}_S(A)|_E$ is ample. Embed E by a power of this line bundle and use its Fubini–Study metric. Lift the embedding sections locally to S and take the appropriate metric root. Their ambient extensions have positive tangential curvature along the smooth part of E; at a node they have positive curvature in all directions, since the Zariski tangent space of E there is the ambient tangent space and the embedding is injective on tangent spaces. Patch these extensions using a partition of unity relative to a reference metric, keeping the chosen metrics near the nodes. The restriction to E, and hence the tangential curvature on its smooth part, is unchanged.

Let $q = \|s_E\|^2$ for any smooth metric on $\mathcal{O}_S(E)$. On the smooth part of E, $dd^c q$ is positive in the normal direction and zero in the tangent direction; at a node it is zero. Adding Cq to the patched weights, for sufficiently large C, makes the curvature strictly positive along all of E, by the Schur complement and compactness, and hence in a neighborhood. For this metric h, $-\log \|s_A\|_h^2$ is the required collar.

If $\lambda = 0$, simplicity gives a one-dimensional positive kernel, which has a rational generator because I is integral. Clearing denominators proves (5.1). Intersections with curves outside E are nonnegative by effectiveness; in the projective case this proves nefness. □

In particular, under the continuous-exhaustion hypothesis a connected SNC boundary is either already handled by a positive collar, or supports the weighted square-zero divisor (5.1). A negative definite boundary is an *exclusion*, not a continuous-versus-smooth counterexample.

5.3. **Nonprojective compactifiers.**

Theorem 5.4 (Nonprojective boundary criterion)**.** *Let S be a smooth compact connected nonprojective surface and let $D \neq 0$ be effective. Then the following are equivalent:*

(i) *$S \setminus \operatorname{supp} D$ has a continuous psh exhaustion;*
(ii) *$S \setminus \operatorname{supp} D$ has a smooth psh exhaustion;*
(iii) *no connected component of $\operatorname{supp} D$ has negative definite intersection matrix.*

No Kähler or SNC hypothesis is required. The Hopf branch of the proof uses the elliptic-neighborhood theorem in the preprint [14, Theorem 1.6].

Proof. Only (iii)⇒(ii) needs proof, by Lemma 5.3(i). We separate the algebraic-dimension cases, using the classification facts in [8, Section 2].

If $a(S) = 1$, the algebraic reduction is a holomorphic elliptic fibration $f : S \to B$ with connected fibers, and every curve is vertical. For a fiber $F = \sum m_i C_i$, its matrix I satisfies

$$x^{\mathsf{T}} I x = -\sum_{i<j} (C_i \cdot C_j) m_i m_j \left(\frac{x_i}{m_i} - \frac{x_j}{m_j} \right)^2 . \tag{5.2}$$

The kernel is the line spanned by (m_i), and every proper principal submatrix is negative definite. Hence the boundary contains every fiber support it meets. Thus $X = f^{-1}(B \setminus P)$ for a finite nonempty set P, and a smooth strictly subharmonic exhaustion of the punctured compact curve pulls back to a smooth psh exhaustion of X. This argument includes reducible and multiple fibers.

Suppose $a(S) = 0$. For every line bundle L on a nonprojective surface, $c_1(L)^2 \leq 0$ [8, Lemma 2.1]. Every boundary matrix is therefore negative semidefinite. Hypothesis (iii) and the same Perron–Frobenius argument give an effective divisor A with positive integral coefficients, full boundary support, and $A^2 = 0$. Contract to a smooth minimal model $p : S \to S_0$. At each blowdown $q : T \to T'$, write $A = q^* A' + bE$, where $A' = q_* A$ and $E^2 = -1$. Then

$$0 = A^2 = (A')^2 - b^2, \qquad (A')^2 \leq 0,$$

so $b = 0$ and $A' \neq 0$. Consequently $A = p^* A_0$, $A_0^2 = 0$, and

$$\operatorname{supp} A = p^{-1}(\operatorname{supp} A_0).$$

The restriction $X \to S_0 \setminus \operatorname{supp} A_0$ is proper. It is enough to construct a smooth exhaustion downstairs.

In the Kähler case, S_0 is a torus or a K3 surface of algebraic dimension zero. A torus of algebraic dimension zero has no curves: an irreducible curve would be elliptic (by adjunction and the absence of rational curves), and its elliptic quotient would give positive algebraic dimension. On a K3 surface, Riemann–Roch gives $h^0(\mathcal{O}(A_0)) \geq 2$ because $A_0^2 = 0$ and $-A_0$ is not effective. Both alternatives contradict $a(S_0) = 0$.

In the non-Kähler case, S_0 is of class VII. If $b_2(S_0) = n > 0$, Enoki's Main Theorem [9, p. 454] identifies a nonzero effective square-zero divisor with a multiple of the distinguished cycle on an Enoki surface. Its complement is an affine-line bundle of degree $-n$ over an elliptic curve [9, (0.4), p. 453]. Complete the affine fibers to a projective $\mathbf{P}^1$-bundle. Its infinity section has square $n > 0$, so Lemma 5.3(iii),(ii) gives a smooth psh exhaustion of the affine-line complement. This uses Enoki's specific square-zero-divisor theorem, not a classification of all class VII surfaces.

If $b_2(S_0) = 0$, a minimal class VII surface containing a curve is elliptic or Hopf [9, p. 453]; algebraic dimension zero leaves Hopf. Each irreducible curve has arithmetic genus one, since $C^2 = K_{S_0} \cdot C = 0$. It cannot have rational

normalization: a map from $\mathbf{P}^1$ lifts to $\mathbb{C}^2 \setminus \{0\}$ and is constant. Thus the curves are smooth elliptic and mutually disjoint. For each of them, finite Ueda type gives a smooth strictly psh divergent collar [13, Theorem 2.5]. Infinite type gives a unitary flat neighborhood by [14, Theorem 1.6], and hence the smooth pluriharmonic collar $-\log|w_i|^2$ in unitary defining functions. Lemma 5.3(ii) finishes the proof. □

In particular, for a nonprojective Kähler compactifier, a nonempty boundary complement can be continuously weakly complete only when $a(S) = 1$ and the boundary is a union of whole fiber supports. The $a(S) = 0$ Kähler branch is excluded, not an unresolved positive case.

5.4. **Smooth rulings.**

Theorem 5.5 (Boundary criterion for a smooth ruling)**.** *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a compact connected Riemann surface, and let $D \neq 0$ be reduced SNC. Then continuous psh exhaustion, smooth psh exhaustion, and the absence of a negative definite connected boundary component are equivalent for $S \setminus D$. There is no restriction on the degree $D \cdot F$.*

Proof. Necessity follows from Lemma 5.3. For sufficiency, positive largest eigenvalue gives a smooth collar. A holomorphic $\mathbf{P}^1$-bundle over a compact curve is a projectivized rank-two bundle; in particular, it is projective and has a section. For a connected component $E = \bigcup E_i$ with zero largest eigenvalue, take $A = \sum a_i E_i$ as in (5.1). We construct a smooth exhaustion of $S \setminus E$, whose restriction supplies the required collar at E.

If every component is vertical, E is one whole fiber, since the fibers of a smooth ruling are irreducible. Pull back an exhaustion of B minus its image. Otherwise $A \cdot F > 0$, so no fiber can be a component of E. Choose a section C_0, with $C_0^2 = -e$; the numerical divisor space has basis C_0, F. Writing $A \equiv aC_0 + bF$, the equality $A^2 = 0$ gives $b = ae/2$. Writing $E_i \equiv d_i C_0 + b_i F$, the identities $A \cdot E_i = 0$ give

$$E_i \equiv d_i(C_0 + (e/2)F), \qquad E_i^2 = E_i \cdot E_j = 0 \quad (i \neq j).$$

Distinct components are therefore disjoint. Connectedness and the SNC hypothesis make $E = C$ a single smooth square-zero curve. For $d = C \cdot F$, adjunction and $K_S \equiv -2C_0 + (2g(B) - 2 - e)F$ give

$$2g(C) - 2 = d(2g(B) - 2).$$

Riemann–Hurwitz shows that $C \to B$ is finite étale.

If $d = 1$, the complement is an affine-line bundle. Its linear bundle is dual to $N_{C/S}$ and therefore has degree zero. We recall the smooth exhaustion construction of Koike–Ueda [15, Proposition 2.1]. In unitary flat frames its transitions are $w_i = a_{ij} w_j + b_{ij}(z)$, $|a_{ij}| = 1$. Choose a smooth section and subtract a global section of the linear bundle so that its $\bar\partial$ is harmonic. On coordinate discs, choose antiholomorphic primitives g_i of this harmonic form.

Changing the affine coordinates by holomorphic translations then gives

$$w_i = a_{ij} w_j + c_{ij}, \qquad g_i = a_{ij} g_j + c_{ij},$$

where c_{ij} is constant: the differences of the g_i are both holomorphic and antiholomorphic. The functions $\rho_i = |w_i - g_i|^2$ agree and have Levi matrix

$$\begin{pmatrix} |(g_i)_{\bar{z}}|^2 & 0 \\ 0 & 1 \end{pmatrix}.$$

They define a smooth psh exhaustion; its sublevels are bounded closed disc bundles around a smooth section over compact B.

For $d \geq 2$, a finite étale Galois base change splits the boundary into d disjoint sections. Each has square zero, since the pullback of C is their disjoint sum and the projection formula applies. If $d = 2$, putting these sections at 0 and ∞ identifies the complement with $L^\times$ for a degree-zero line bundle. A unitary flat metric makes $(\log \|v\|^2)^2$ a smooth psh exhaustion. If $d \geq 3$, three labeled sections trivialize the $\mathbf{P}^1$-bundle; the others are constant, as maps from compact B' to $\mathbf{P}^1$ avoiding $0, 1, \infty$. The complement is $B' \times (\mathbf{P}^1 \setminus Q)$ for a finite set Q, and a smooth subharmonic exhaustion of the punctured projective line pulls back. Average a nonnegative exhaustion over the finite deck group. The average remains proper, being at least the original exhaustion divided by the group order, and descends smoothly to $S \setminus C$. Finally glue the collars at all connected boundary components using Lemma 5.3(ii). □

This theorem uses the actual smooth ruling, not merely birational ruledness. It does not claim the same result for arbitrary blowups of a ruled surface or for singular rational fibrations. For $S = \mathbf{P}^2$, every nonzero effective divisor is ample, so its complement has a smooth psh exhaustion directly from a positive metric on $\mathcal{O}_S(D)$.

5.5. **Further projective cases and the numerical residual class.**

Proposition 5.6 (Projective affirmative cases)**.** *Let S be smooth projective, let $D \neq 0$ be reduced SNC, and assume $X = S \setminus D$ has a continuous psh exhaustion. Then X has a smooth psh exhaustion in each of the following cases:*

(i) *D is disconnected;*
(ii) *on every connected boundary component with zero largest intersection eigenvalue, a positive integral kernel divisor A satisfies $\kappa(S, A) = 1$ or $K_S \cdot A < 0$;*
(iii) *K_S is torsion in* $\mathrm{Pic}(S)$*;*
(iv) *every component of D is vertical for a morphism $f : S \to B$ with connected fibers onto a smooth projective curve.*

Proof. For (i), write the connected boundary components as $E_1, \ldots, E_r$, $r \geq 2$. The matrix argument provides effective A_k with support E_k and $A_k^2 \geq 0$, with strict inequality in the positive-eigenvalue case. Since $A_k \cdot A_l = 0$

for $k \neq l$, the Hodge index theorem excludes a positive square and makes the nonzero square-zero nef classes proportional. The proportionality factors are positive rational numbers, by intersection with an integral ample class. Choose positive integers p_k, q_k such that $p_k A_k - q_k A_1$ is numerically trivial, for $k \geq 2$. Its real first Chern class is zero by Hodge index, and the $\partial\bar{\partial}$-lemma gives a unitary flat metric h_k on $\mathcal{O}_S(p_k A_k - q_k A_1)$. For its canonical meromorphic section σ_k, the functions

$$\phi_k = \log \|\sigma_k\|^2_{h_k}, \qquad \rho = \sum_{k=2}^{r} \phi_k^2$$

are smooth on X; the ϕ_k are pluriharmonic, so ρ is psh. It tends to infinity at every boundary component, proving properness.

For (ii), a nef divisor of Iitaka dimension one on a surface is semiample [11, Lemma 7.3]. If $K_S \cdot A < 0$, Riemann–Roch gives

$$\chi(\mathcal{O}_S(nA)) = \chi(\mathcal{O}_S) - \tfrac{n}{2} K_S \cdot A \longrightarrow +\infty.$$

Serre duality and intersection with an ample class give $H^2(S, \mathcal{O}_S(nA)) = 0$ for large n. Thus $\kappa(S, A) \geq 1$, and nefness with $A^2 = 0$ gives $\kappa(S, A) = 1$. In either case A is semiample. A base-point-free multiple defines, after Stein factorization, a morphism $f : S \to B$ to a smooth curve with connected fibers and $\mathcal{O}_S(mA) = f^*L$. The canonical section descends because $f_*\mathcal{O}_S = \mathcal{O}_B$, so $mA = f^*G$ for a nonzero effective divisor G on B. Hence

$$S \setminus \operatorname{supp} A = f^{-1}(B \setminus \operatorname{supp} G),$$

with proper restricted map. Pulling back an exhaustion of the punctured curve gives a collar at $\operatorname{supp} A$. Combine these with the positive-eigenvalue collars using Lemma 5.3(ii).

For (iii), consider such a square-zero nef A. Choose a small positive rational ε so that every coefficient of εA is at most one. The log divisor $K_S + \varepsilon A$ is nef. Log abundance for surfaces [10, Theorem 6.1] makes it semiample. Killing the torsion of K_S and clearing denominators shows that A itself is semiample. Apply the preceding fibration argument. This is a direct consequence of log abundance on the given pair; no change of the open surface by a log minimal model program is involved.

For (iv), identity (5.2) holds for the multiplicity vector of any fiber. Every proper subset of its components has negative definite intersection matrix. Therefore the boundary obstruction forces D to be a union of whole fiber supports. The restricted map $X \to B \setminus P$ is proper for a finite nonempty set P; pull back an exhaustion of the punctured base. □

In (i) no SNC assumption is necessary, because boundary resolution preserves the complement and its number of boundary components. Part (iii) applies, in particular, to the minimal projective surfaces of Kodaira dimension zero: K3, Enriques, abelian, and bielliptic surfaces [7, Chapter VI]. This is a standard consequence of the cited abundance theorem, not a new

classification result. Part (iv) includes the vertical-boundary case of an elliptic fibration, but does not treat arbitrary horizontal boundaries.

Corollary 5.7 (Residual projective pairs)**.** *If a divisor complement in a smooth projective surface has a continuous psh exhaustion but no smooth psh exhaustion, then after boundary resolution it must have connected reduced SNC boundary D supporting*

$$A = \sum_i a_i D_i, \quad a_i \in \mathbb{Z}_{>0}, \quad A \text{ nef}, \quad A \cdot D_i = 0, \quad A^2 = 0,$$

$$\kappa(S, A) = 0, \qquad K_S \cdot A \geq 0.$$

The continuous-exhaustion hypothesis is retained; these numerical conditions alone assert neither its existence nor failure of smooth exhaustion.

Proof. Disconnected boundary and the positive-eigenvalue case are already settled. In the remaining case the matrix lemma gives A. It is effective, so $\kappa(S, A) \geq 0$, and a nef square-zero divisor has Iitaka dimension at most one. Proposition 5.6 excludes dimension one and $K_S \cdot A < 0$. □

5.6. **A concise smooth-boundary criterion.** For a smooth square-zero curve C, its normal bundle N is unitary flat. The Ueda type measures the orders to which local defining functions can be made to have unitary linear transition functions. Infinite type is formal; a unitary flat neighborhood requires *convergent* defining functions $w_i = t_{ij} w_j$, $|t_{ij}| = 1$. These two conditions must not be conflated [12, 13].

Proposition 5.8 (Sufficient conditions, not a classification)**.** *Let S be a compact Kähler surface and let $C \subset S$ be smooth, connected, and square-zero. Write $\rho : H_1(C, \mathbb{Z}) \to \mathrm{U}(1)$ for the unitary character of $N_{C/S}$ and $K = \ker[H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})]$. Then $S \setminus C$ has a smooth psh exhaustion if any of the following holds: finite Ueda type; a unitary flat neighborhood; torsion or Diophantine normal bundle; $\rho(K)$ finite; $q(S) \geq g(C)$; or $g(C) = 1$. No continuous exhaustion is assumed. Only the last case uses* [14, Theorem 1.6].

Proof. Finite type gives a smooth strictly psh collar [13, Theorem 2.5]. A flat neighborhood gives the pluriharmonic collar $-\log|w_i|^2$. In infinite type, torsion or Diophantine N gives such a neighborhood by Ueda's convergence theorem [12, Theorem 3]; finite type has already been handled. Here Diophantine means $\mathrm{dist}(N^m, \mathcal{O}_C) \geq c m^{-\tau}$ for some $c, \tau > 0$ and all $m \geq 1$.

If $\rho(K)$ is finite, a power ρ^k is trivial on K, hence defines a character of the image of $H_1(C, \mathbb{Z})$ in $H_1(S, \mathbb{Z})$. It extends to the whole group because $\mathrm{U}(1)$ is divisible. In the remaining non-torsion, infinite-type case, Neeman's extension theorem in [13, Theorem 4.4] gives an effective divisor E disjoint from C and cohomologous to mC. A unitary flat metric on $\mathcal{O}_S(mC - E)$ then gives the pluriharmonic collar $-m^{-1}\log\|\sigma\|^2$ at C.

For $q(S) \geq g(C)$, consider $r : H^1(S, \mathcal{O}_S) \to H^1(C, \mathcal{O}_C)$. If r is not injective, [13, Proposition 2.7] makes a multiple of C a fiber of a fibration,

and the punctured-base exhaustion applies. Otherwise the dimension inequality makes r an isomorphism. Hodge decomposition and duality make $H_1(C,\mathbb{Q}) \to H_1(S,\mathbb{Q})$ an isomorphism. Its integral kernel is torsion and lies in the free abelian group $H_1(C,\mathbb{Z})$, so $K = 0$. Apply the character case. Finally, when $g(C) = 1$, infinite type implies a unitary flat neighborhood by [14, Theorem 1.6]. In every case finish with Lemma 5.3(ii). □

The finite-type and flat-neighborhood arguments also apply without a Kähler hypothesis, as used in the Hopf branch above. Neither semipositivity of $\mathcal{O}_S(C)$ nor holomorphic convexity of $S \setminus C$ is asserted as a necessary condition: the proposition constructs collars, not a general equivalence with either of those properties.

5.7. **What the minimal-surface classification does and does not give.** In Table 1, S is a *smooth compact minimal* surface (no smooth rational (-1)-curve), $D \neq 0$ is reduced SNC, and $X = S \setminus D$ is assumed to have a continuous psh exhaustion. "Affirmative" means that the preceding arguments give a smooth one; "excluded" means the hypotheses cannot occur; "partial" means that only the stated additional conditions have been handled. The table uses the Enriques–Kodaira classification [7, Chapter VI], but every exhaustion assertion is justified by the results above. It is not a classification of noncompact surfaces or a claim that every partial case is open in the literature.

The nonprojective theorem already includes nonminimal compactifiers, by its full-pullback argument. The projective rows do *not* automatically extend to all surfaces birational to the listed minimal models. In particular, the table does not settle arbitrary rational surfaces obtained by boundary-changing blowups of $\mathbf{P}^2$ or a ruling. Conversely, an open surface need not be presented by a compact minimal pair at all. The phrase "only $\kappa = 1, 2$ remain" would therefore be incorrect for the unrestricted surface problem.

For a smooth connected square-zero boundary on a projective surface, any putative counterexample must avoid every condition in Proposition 5.8; in particular it must have $g(C) \geq 2$, $q(S) < g(C)$, infinite Ueda type without a unitary flat neighborhood, and $\rho(K)$ infinite. These are necessary conditions, not examples. The remaining analytic task is to decide whether a continuous divergent psh collar in such a neighborhood forces the existence of a smooth one, or to construct a pair where it does not. The numerical residual conditions do not answer that task.

The general continuous-to-C^2 implication for complex surfaces is not resolved by this paper, even under Kählerness or for all smooth quasi-projective surfaces. The positive statements in this section, including the explicitly identified recent elliptic input, are independent of the threefold counterexamples.

TABLE 1. Divisor complements with a minimal compactifier; a continuous psh exhaustion is assumed.

Compactifier S	Exhaustion conclusion and exact scope
Projective, $\kappa(S) = -\infty$: $\mathbf{P}^2$ or a minimal ruled surface	**Affirmative.** Ample boundary in $\mathbf{P}^2$; the smooth-ruling criterion otherwise (Theorem 5.5).
Projective, $\kappa(S) = 0$: K3, Enriques, abelian, bielliptic	**Affirmative.** The canonical bundle is torsion; Proposition 5.6(iii) applies.
Projective, $\kappa(S) = 1$: properly elliptic	**Partial.** Vertical boundary is handled. Horizontal boundary requires another stated criterion; no blanket conclusion is proved.
Projective, $\kappa(S) = 2$: general type	**Partial.** Disconnected, positive-collar, moving, or the specified smooth-boundary cases are handled. The residual connected square-zero case is not settled.
Nonprojective, $a(S) = 1$	**Affirmative.** Continuous exhaustion forces the boundary to be a union of whole elliptic fiber supports. This includes the relevant Kodaira and elliptic surfaces.
Nonprojective Kähler, $a(S) = 0$: torus or K3	**Excluded.** No nonempty divisor complement here satisfies the continuous-exhaustion hypothesis.
Nonprojective, $a(S) = 0$, class VII, $b_2 = 0$	**Affirmative when a boundary exists.** Such a surface is Hopf; elliptic collars apply, using [14]. Curve-free surfaces have no nonempty divisor boundary.
Nonprojective, $a(S) = 0$, class VII, $b_2 > 0$	**Affirmative for every admissible boundary.** The matrix condition forces the Enoki square-zero-divisor case. All other boundaries are excluded by the negative definite obstruction.

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